Analysis of Bias in Anime Recommendation Algorithms

Damian Gubbins
Departamento de Ciencias de la Computación
Facultad de Ingeniería
Pontificia Universidad Católica de Chile
damiangubbins@uc.cl

Jorge de la Fuente
Departamento de Ciencias de la Computación
Facultad de Ingeniería
Pontificia Universidad Católica de Chile
jorge.delafuente@uc.cl

Nicolás Fernández
Departamento de Ciencias de la Computación
Facultad de Ingeniería
Pontificia Universidad Católica de Chile
nifernandez@uc.cl

Benjamín Huenchuñir
Departamento de Ciencias de la Computación
Facultad de Ingeniería
Pontificia Universidad Católica de Chile
bahuenchunir@uc.cl

Abstract

Japanese animated series and films, known as anime, have become a global phenomenon and represent a rapidly expanding market expected to continue growing in the coming years. As with other sectors in the entertainment industry, this growth intensifies the challenge of delivering accurate and meaningful recommendations—especially given the vast variety of anime categories that appeal to a diverse user base.

Within the context of recommendation systems, an important ethical concern arises: how bias is created, reinforced, and handled. Biased recommendations can lead not only to poor or inaccurate suggestions but also to discriminatory outcomes for certain users based on their characteristics.

In this paper, we aim to study and evaluate models that can provide strong predictive performance while minimizing bias. By examining available datasets and the influence of model selection and fitting strategies, we seek to identify methods that reduce the bias present in recommendation outputs.

1 Introduction

In recent years, Japanese animated series and films (anime) have become increasingly popular among young audiences. They have not only shaped generations in Japan, but have also spread worldwide, showcasing the culture and values characteristic of the Asian country. According to Grand View Research (n.d.), "The global anime market was valued at USD 34,256.2 million in 2024 and is projected to grow to USD 60,272.2 million by 2030" (author's translation, first paragraph), which implies that the market will double in the coming years.

Anime is divided into a wide variety of categories depending on the type of story, and there are options for all kinds of viewers. Because of this wide range of choices, users often do not know what to watch, or feel overwhelmed by the number of options and wish to find the one that best fits their preferences.

Our goal is not to design a new recommendation system, but rather to employ several existing models and analyze the potential recommendation biases that may arise in relation to the user's gender,

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age, and location. This will allow us to determine the kinds of anime that are recommended more frequently to certain types of users. In other words, our work focuses on a fairness analysis.

Bias in recommender systems is a very important factor to consider. While these systems aim to maximize user utility, inherent biases in training data and algorithmic design can lead to two major categories of harm: utility harm (resulting in poor recommendations for specific subgroups) and allocational or representational harm (which can discriminate against individuals based on protected characteristics such as gender, age, or ethnicity). Consequently, a fundamental challenge within the field of AI is the development of robust methodologies to accurately analyze, quantify, and mitigate the disparate impact present in recommendation algorithms. In 2021, during a UNESCO conference on AI, this issue was addressed: “AI technologies can be of great service to humanity and all countries can benefit from them, but also raise fundamental ethical concerns, for instance regarding the biases they can embed and exacerbate, potentially resulting in discrimination, inequality, digital divides, exclusion and a threat to cultural, social and biological diversity” (UNESCO, 2021). This perfectly illustrates the problem, recommender system without bias mitigation can discriminate, which is detrimental for the system success in the market and to its users.

For these reasons, we believe that studying this bias factor across different models is both interesting and highly relevant, we will focus on analyzing gender and age bias, as these appear to be the most common in anime recommender systems.

2 Related Work

The development of recommendation systems has been a central topic in machine learning research for decades. In the specific context of anime, approaches have evolved from traditional filtering methods to complex deep learning architectures. However, while predictive accuracy has improved, recent literature highlights a critical need to address the ethical implications and biases inherent in these algorithms.

2.1 Anime Recommendation Systems

Early research in the anime domain predominantly relied on Collaborative Filtering (CF) and Content-Based Filtering (CBF). Collaborative filtering, particularly Matrix Factorization techniques like Alternating Least Squares (ALS), has been widely adopted due to its ability to capture latent user preferences without requiring extensive metadata (Koren et al., 2009). These models rely on the user-item interaction matrix, which, as noted in our dataset analysis, often suffers from high sparsity in the anime domain.

Conversely, Content-Based approaches leverage the rich metadata available for anime, such as genres, studios, and synopses. Recent studies have demonstrated that calculating cosine similarity on vector representations of anime descriptions can effectively mitigate the "cold start" problem for new items (Balaji et al., 2023). However, these systems often suffer from over-specialization, creating "filter bubbles" where users are only recommended items very similar to what they have already seen, limiting diversity.

2.2 Deep Learning and Factorization Machines

To overcome the limitations of linear models, the field has shifted toward Deep Learning. Neural Collaborative Filtering (NCF) and Factorization Machines (FM) have become state-of-the-art baselines. Specifically, DeepFM, proposed by Guo et al. (2017), represents a significant advancement by combining Factorization Machines (which model low-order feature interactions) with Deep Neural Networks (which capture high-order, non-linear interactions) in an end-to-end architecture.

DeepFM is particularly relevant for datasets with both sparse categorical data (like user IDs) and dense features, as it avoids the need for extensive manual feature engineering. While DeepFM typically achieves higher accuracy metrics (Recall, AUC) compared to traditional methods, its complexity raises questions about interpretability and the potential amplification of training data biases.

2.3 Bias and Fairness in Recommender Systems

As recommendation algorithms become ubiquitous, the research focus has expanded from pure accuracy to fairness and ethics. Bias in recommender systems is typically categorized into data bias and algorithmic bias.

Popularity Bias is the most well-documented issue, where algorithms disproportionately recommend popular items (the "short head") while suppressing niche items (the "long tail"). Abdollahpouri et al. (2019) demonstrated that this bias not only hurts item discovery but can also be unfair to users with niche tastes, as the system fails to deliver relevant recommendations to them.

More critically, Demographic Bias (or Disparate Impact) occurs when a model performs significantly worse for specific protected groups, such as gender or age. Ekstrand et al. (2018) highlighted that standard collaborative filtering algorithms often yield lower prediction accuracy for minority demographic groups. This disparity arises because the majority group dominates the training signal, causing the model to optimize its parameters primarily for that group.

Our work builds upon these foundations by applying the Statistical Parity framework to the specific domain of anime. While previous studies have analyzed fairness in general movie datasets (e.g., MovieLens), there is a gap in the literature regarding how these biases manifest in the anime market, which is characterized by distinct demographic skews and consumption patterns. We aim to evaluate whether advanced models like DeepFM mitigate or exacerbate these biases compared to traditional baselines.

3 Dataset and Baseline Assessment

The analysis began with an exploratory data analysis of the initial dataset, which revealed three critical limitations that motivated the subsequent data enrichment process:

- **Data Scarcity (Cold Start):** The user–item matrix exhibited extreme sparsity, with a density of approximately 0.07% in the training set. This resulted in an average of fewer than 2 ratings per user, severely limiting the applicability and performance of traditional Collaborative Filtering methods.
- **Rating Bias:** The rating distribution showed a strong positive skew, with over 60% of ratings concentrated at the high end (scores of 8, 9, and 10), potentially leading to inflation in predicted scores.
- **Demographic Bias:** The user profiles displayed a strong demographic skew, with male users representing approximately 70% of the population and the majority of users concentrated among young adults (27–32 years old). This imbalance presents a significant risk of recommendation bias against minority subgroups. (The age distribution is detailed in Figure 1).

Baseline Model Evaluation

To formally quantify the impact of the data scarcity and popularity bias, we implemented two fundamental baseline models: **Random** and **Most Popular (MP)**. The comparative results, summarized in Table 1, confirm two distinct trade-offs:

1. The MP model significantly outperformed Random across all accuracy metrics (Recall, MAP, nDCG), demonstrating the effectiveness of popularity-based heuristics in sparse environments.
2. Conversely, the MP model exhibited extremely low **Novelty** and **Diversity** (a score of $\mathbf{0.843}$), confirming that it primarily recommends a small, non-diverse set of well-known anime. This result underscores the direct link between popularity-based algorithms and limited content exposure.

Dataset Enrichment

To mitigate the **cold start** and **sparsity** issues, a targeted data enrichment process was performed. This involved a significant expansion of the original dataset through additional scraping from

Table 1: Comparative baseline results at $K = 10$

Metric	Random	Most Popular (MP)
Recall@K	~ 0.0003	$\mathbf{0.00450}$
MAP@K	~ 0.0001	$\mathbf{0.00182}$
nDCG@K	~ 0.0002	$\mathbf{0.00250}$
Novelty	$\sim \mathbf{7.1}$	4.52860
Diversity	$\sim \mathbf{1.0}$	0.84311

MyAnimeList, focusing on obtaining more recent ratings and capturing users with longer review histories.

This scraping process resulted in a substantial increase in the average number of ratings per user, enriching the contextual information available for both items and users. The success of this data expansion is clearly illustrated by the post-expansion distribution shown in Figure 3, which contrasts sharply with the extreme sparsity of the initial dataset. This improved density was crucial for the successful application and analysis of subsequent collaborative filtering models.

Table 2: Statistics of the enriched dataset

Statistic	Training	Validation
Number of users	28196	15089
Number of anime	7156	4488
Total ratings	139568	33653
Average ratings per user	4.95	2.23
Average ratings per anime	19.50	7.50
Mean rating	7.60	7.62
Rating standard deviation	2.20	2.18
Highest number of ratings by a user	651	147
Highest number of ratings for an anime	1239	314
Density (%)	0.0692%	0.0497%

As shown in Table 2, the scraping results have enabled a **substantial improvement** of the interaction matrix, directly addressing the cold start and data scarcity problem identified initially. The **density has increased from** 0.07% **to** $\mathbf{0.0738}\%$. Although the percentage increase may appear small, the impact is significant because the **number of users** ($\sim 28,400$) has quadrupled and the **total ratings** ($\sim 151,000$) in Training have doubled. Crucially, the **average ratings per anime have increased from** ~ 4 **to** $\sim \mathbf{21}$, providing more contextual information per item. Furthermore, the presence of users with up to **748 ratings** in Training, and anime with up to **1219 ratings**, confirms that the new strategy has effectively captured more active users.

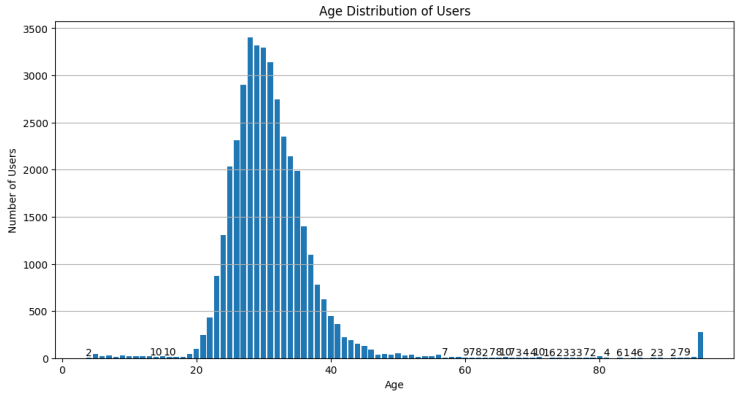

Figure 1: Age distribution of users in the dataset.

4 Methodology Used

For the analysis, three recommendation models were evaluated: ALS, Anurag122002, and DeepFM:

Model 1: ALS is a collaborative filtering model based on matrix factorization.

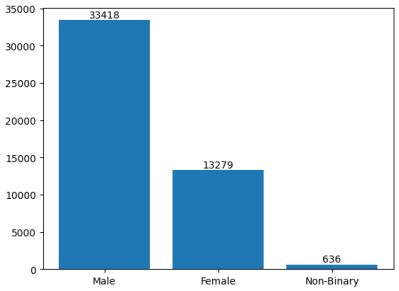

Figure 2: Gender distribution of users in the dataset.

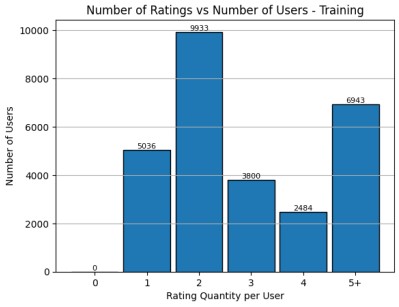

Figure 3: Number of ratings vs. number of users (Training).

Model 2 (Anurag122002) uses an approach based exclusively on the content of the anime, ignoring user interactions and ratings. This system generates vector representations from metadata such as descriptions and genres, and then recommends similar items using cosine similarity.

Identified biases:

- **Serialization bias:** The model prioritizes anime with multiple seasons, movies, or derivative content, since their larger number of attributes increases the probability of appearing in recommendations. This limits diversity by favoring established franchises over standalone or lesser-known works.
- **Lack of social context:** By not incorporating user ratings or reviews, the system omits key dimensions of collective preference, such as perceived quality or temporal trends. This can produce recommendations that are technically similar but perceptually irrelevant for users.
- **Diversity problem:** When comparing vectors based on synopses, the model tends to recommend anime, movies, and OVAs from the same franchise because their synopses are highly similar. This may lead to recommendations that are not very diverse and highly similar to each other.

Model 3: DeepFM is a hybrid Deep Learning model that combines factorization machines and deep neural networks.

The main evaluation metric used is the **Statistical Parity Ratio**, which is calculated from metrics such as Recall@k, Precision@k, MAP@K, nDCG@k, Novelty, and Diversity.

5 Parameter Analysis

To classify whether a model exhibits **Disparate Impact** bias, we quantify the difference in performance between two user subsets, *A* and *B*, using the **Statistical Parity Ratio (SPR)** for a given metric M. The SPR, denoted as Ratio(M), is modeled by the following formula:

$$\text{Ratio}(M) \;=\; \frac{\min\left(M_{\text{A}},\, M_{\text{B}}\right)}{\max\left(M_{\text{A}},\, M_{\text{B}}\right)} \tag{1}$$

The value Ratio(M) approaches 1.0 when there is no significant performance difference between the groups. We define the **Fairness Ratio Threshold** as $T = 1 - \alpha$, where α represents the accepted **Disparity Margin** (e.g., $\alpha = 0.2$). If Ratio$(M) < T$, the model is considered biased for the metric M under evaluation.

This analysis is primarily applied for the comparison of protected gender subsets (Male vs. Female), but the methodology is equally applicable to any two demographic or behavioral subgroups, such as different age ranges (e.g., 25–34 vs. 45–54).

Furthermore, we also analyzed the **Disparity in Category Exposure** between the Male and Female subsets by comparing the proportion of anime of a certain genre presented to each group. This

allows for the detection of **Stereotypical Bias**, revealing whether the recommendation algorithm disproportionately allocates different types of content (genres) to each group, thus reinforcing or amplifying existing societal norms and potentially limiting the content diversity offered to a specific gender.

6 Results

A detailed summary of the disparity analysis is presented in Tables 3 to 9.

Table 3: ALS (Female vs. Male)

Metric	**Ratio**	**Biased** (≤ 0.80)	**Favored Group**
Recall	0.665	**Yes**	Male
Precision	0.651	**Yes**	Male
MAP	0.714	**Yes**	Male
nDCG	0.686	**Yes**	Male
Novelty	0.994	No	Male
Diversity	0.998	No	Male

Table 4: ALS (25–34 vs. 45–54)

Metric	**Ratio**	**Biased** (≤ 0.80)	**Favored Group**
Recall	0.271	**Yes**	25–34
Precision	0.252	**Yes**	25–34
MAP	0.313	**Yes**	25–34
nDCG	0.300	**Yes**	25–34
Novelty	0.997	No	25–34
Diversity	0.995	No	45–54

Table 5: Anurag122002 (Female vs. Male)

Metric	**Ratio**	**Biased** (≤ 0.80)	**Favored Group**
Recall	0.681	**Yes**	Male
Precision	0.679	**Yes**	Male
MAP	0.702	**Yes**	Male
nDCG	0.693	**Yes**	Male
Novelty	0.986	No	Male
Diversity	0.987	No	Female

Table 6: Anurag122002 (25–34 vs. 45–54)

Metric	**Ratio**	**Biased** (≤ 0.80)	**Favored Group**
Recall	0.768	**Yes**	25–34
Precision	0.832	No	25–34
MAP	0.480	**Yes**	25–34
nDCG	0.571	**Yes**	25–34
Novelty	0.980	No	45–54
Diversity	0.961	No	45–54

Table 7: DeepFM (Female vs. Male)

Metric	**Ratio**	**Biased** (≤ 0.80)	**Favored Group**
Recall	0.842	No	Male
Precision	0.798	**Yes**	Male
MAP	0.934	No	Male
nDCG	0.887	No	Male
Novelty	0.964	No	Female
Diversity	0.996	No	Male

Table 8: DeepFM (25–34 vs. 45–54)

Metric	**Ratio**	**Biased** (≤ 0.80)	**Favored Group**
Recall	0.787	**Yes**	25–34
Precision	0.769	**Yes**	25–34
MAP	0.378	**Yes**	25–34
nDCG	0.572	**Yes**	25–34
Novelty	0.846	No	45–54
Diversity	0.971	No	45–54

Table 9: Categorical Exposure Disparity in Anurag122002 (Female vs. Male)

Category	**Ratio**	**Biased** (≤ 0.80)	**Favored Group**
More Biased towards Female			
Shoujo	0.549	**Yes**	Female
Demons	0.597	**Yes**	Female
Dementia	0.611	**Yes**	Female
Shounen Ai	0.659	**Yes**	Female
Josei	0.759	**Yes**	Female
Historical	0.746	**Yes**	Female
More Biased towards Male			
Cars	0.502	**Yes**	Male
Space	0.627	**Yes**	Male
Game	0.767	**Yes**	Male
Mecha	0.775	**Yes**	Male
More Balanced (Less Biased)			
Shounen	0.997	No	Female
Mystery	0.969	No	Female
Parody	0.933	No	Female
Supernatural	0.933	No	Female
Music	0.932	No	Female
Drama	0.950	No	Female
Magic	0.943	No	Male
Horror	0.862	No	Female
Action	0.860	No	Male

7 Discussion

The models under study showed varying performance. ALS and Anurag122002 exhibited a **strong presence of gender bias**, while DeepFM showed a **moderate presence of age bias**.

Regarding gender bias, the disparity in model performance is heavily influenced by the underlying class imbalance, where the dataset contains 33,000 male users versus only 14,000 female users (an approximate 2.4:1 ratio).

ALS and Anurag122002 prove highly sensitive to this imbalance, exhibiting a strong bias toward the majority male group. Both models fail to reach the fairness threshold across all accuracy metrics, with parity ratios fluctuating between 0.65 and 0.71. This indicates that female users receive recommendations that are roughly 30–35% less accurate than their male counterparts.

Beyond quantitative metrics, Anurag122002 exhibits a behavior based on stereotypes. Table 9 illustrates this: male users receive a higher proportion of recommendations for genres such as "Cars" or "Mecha," while female users more frequently receive recommendations for "Shoujo." This indicates that the model is reinforcing stereotypes based on the user's gender, which can not only reduce the quality of the recommendation but also introduce a form of gender discrimination. On the other hand, genres such as "Shounen," originally created for a male audience, do not show an overrecommendation toward male users, which is a positive outcome in terms of bias. However, this may be explained by the high popularity of anime within this genre, which causes the user's gender to be less relevant when generating recommendations.

In contrast, DeepFM demonstrates remarkable robustness. Despite training on the same imbalanced data, it achieved near-parity in ranking metrics (MAP 0.934 and nDCG 0.887). While it showed a marginal dip in Precision (0.798), DeepFM effectively mitigated the popularity bias that plagued the other architectures. This suggests that DeepFM's ability to model high-order feature interactions allows it to generalize user preferences better than standard matrix factorization approaches, making it the only ethically viable candidate for deployment among the three.

With respect to age bias, all three models consistently favor the younger demographic (25–34), revealing a systemic failure in serving the older group (45–54). While DeepFM and Anurag122002 exhibit strong bias, ALS demonstrates the most severe disparity, with ratios plummeting to ≈ 0.27 across accuracy metrics.

This performance degradation is directly attributable to the significant data imbalance present in the dataset. As observed in the age distribution, the 25–34 age group constitutes the dominant majority, while the 45–54 group represents a sparse minority. Consequently, the models overfit to the preferences of the younger cohort. This is most evident in the ranking metrics (MAP and nDCG), where the models fail to surface relevant items for older users, likely defaulting to items that are globally popular among the younger majority. Although Anurag122002 shows slight robustness in Precision (0.832), the universal collapse in ranking metrics confirms that none of the architectures effectively mitigate this population imbalance without explicit intervention.

Finally, the Novelty and Diversity metrics did not show bias in any case (ratios close to 1.0). Since these metrics evaluate item information rather than user information, the distribution of "novelty" is democratic for all groups.

8 Conclusion

The results show that the strong overall performance of the models hides significant disparities. While some models exhibit strong gender bias, others penalize item ranking based on age. This confirms that optimizing traditional metrics does not guarantee fairness, resulting in systems that systematically favor dominant demographic groups.

The issue of bias arises primarily from two factors: the predominance of male-generated reviews and the limitations of our current recommendation methodology. By relying on a binary recommendation model that simply predicts which items a user might like, we restrict ourselves to an incomplete understanding of user preferences. Moving beyond this paradigm toward a more nuanced approach, one that focuses not merely on what a user may like, but on what they may genuinely want to experience next, could significantly reduce the bias embedded in data-driven recommendations.

Developing systems capable of capturing deeper user intent, rather than surface-level preferences, offers a promising path toward mitigating these biases.

Finally, having a more balanced dataset in terms of the number of male and female users would allow for an even more accurate analysis than the one presented here. However, the amount of data used in this study is sufficiently large to draw robust conclusions.

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